Natural scale selection in washover deposit formation during extreme events

Spectral estimators and XBeach simulations
of the Castelldefels beach

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Abstract

During storms that overtop a beach berm, XBeach simulations of Castelldefels beach (Barcelona) produce washover deposits that self-organize into quasi-periodic channels, whose spacing coarsens through discrete fusions and settles at a late value λ_∞ of 200–350 m. This report studies what selects its value. In 158 surfbeat simulations, each configuration run with three initial perturbations, the still-water level, the wave height and period, the bed friction and the bed-slope coefficient `facsl` are varied. The spacing is measured with the Fourier spectral centroid of a crest signal and with a wavelet centroid. The maximum of the wavelet spectrum is discarded because it locks to the modes of the periodic domain. A conclusion is retained only if it holds with both estimators. The diffusion–incision balance of Perron et al. (2009), with `facsl` as diffusivity, predicts that λ_∞ grows as $\texttt{facsl}^{1/3}$; the measured exponent is -0.08 ± 0.03 (Fourier) and -0.03 ± 0.04 (wavelet). Comparing runs at equal overwash volume rather than at equal storm time does not reduce the spread of the spacing along any parameter axis, so the overwash volume does not explain the effect of the parameters on the spacing. Wave height, wave period and water level increase λ_∞ (exponents $+0.4$ to $+0.6$ with the Fourier estimator, same sign but not significant with the wavelet estimator), which is consistent with the hydrodynamic feedback described by Carrion-Bertran et al. (2026). Changing the domain width from 1000 to 2400 m changes λ_∞ by about 15 %, comparable to the variability between initial perturbations.

1 Introduction

During extreme storms, the combination of surge, setup and wave runup can exceed the crest of a beach berm or barrier, and water and sediment are transported landward over it. This process, overwash, leaves sediment bodies on the subaerial beach known as washover deposits. Its morphological impact depends on how the water level compares with the elevation of the beach crest: Sallenger (2000) classified storm impacts on barriers into swash, collision, overwash and inundation regimes according to the elevation reached by the water, including runup, relative to the base and the crest of the first line of defence (dune or berm). Castelldefels beach, south of Barcelona, is a low-lying Mediterranean beach where washover deposits can form under extreme storms; overwash was observed there with land-based video cameras during the storm of December 2019 (Carrion-Bertran et al., 2026).

Washover deposits are often not uniform alongshore: rhythmic sequences of deposits separated by narrow incising channels have been reported in the field and in the laboratory (Lazarus and Armstrong, 2015). Numerical simulations of the Castelldefels beach show the same behaviour: starting from an alongshore-uniform topography, the deposits self-organize into quasi-periodic channels, and the final deposited volume depends strongly on whether the wavelength of an initial alongshore pattern matches or exceeds this natural scale (Carrion-Bertran et al., 2026). In the simulations analysed here, the channels first appear at short spacings and then coarsen through discrete fusions until a late, quasi-steady spacing λ_∞ is reached.

Since the self-organization is already established, the focus is on what selects λ_∞ . We study how the emergent spacing responds to the physical parameters of the problem, and which of the candidate mechanisms proposed in the literature agrees with that response. Section 2 reviews the two studies that motivate the work; Sections 3 and 4 describe the model and the simulation campaign; Section 5 describes the estimators of the spacing and how the wavelet estimator was adapted to the periodic domain; Section 6 states the candidate mechanisms and their predictions; Section 8 presents the results, which are discussed in Section 9.

2 Motivation: self-organization in washover deposits

2.1 Carrión-Bertrán et al. 2026

Carrion-Bertran et al. (2026) used the 2DH XBeach model in surfbeat mode to study how the initial topobathymetry of Castelldefels beach conditions the formation of washover deposits during an extreme storm. Starting from alongshore-uniform profiles, the deposits became periodic alongshore through a self-organization process: water overtopping the berm decelerates over incipient deposits, which favours further deposition, while the return flow accelerates between them and incises channels. The spacing reached in this way, the natural wavelength of the beach, was 250–300 m for most profiles (about 400 m for one of them), and was measured by detecting the channels as valleys of the alongshore bed profile. When an initial pattern (megacusps or rhythmic dunes) had a wavelength equal to or larger than this natural one, the deposits locked onto it from the beginning and the deposited volume increased, by up to a factor of four for megacusps and less for rhythmic dunes. Each case was simulated with nine realizations of the wave-group chronology (`random=1`) to arrive at a reasonable expected value and its uncertainty, since the variability in each realization was of the same order as the effects being studied.

Similar self-organized patterns are known in other coastal settings, such as beach cusps and crescentic or transverse bars.

2.2 Lazarus & Armstrong 2015

Quasi-periodic washover patterns have been observed along barrier coastlines, and have traditionally been attributed to a forcing template: a periodic structure in the nearshore bathymetry, in the pre-storm topography, or in the hydrodynamic forcing itself, which is then imprinted on the deposits (Lazarus and Armstrong, 2015). Lazarus and Armstrong (2015) proposed an

alternative. In a physical experiment, complemented by a numerical model, they showed that quasi-periodic washover patterns can self-organize, arising from interactions between barrier topography, the routing of overwash flow and sediment flux. They identified the dominant spacing from a wavelet analysis of the back-barrier shoreline.

Both studies thus show that alongshore-periodic washover deposits can form without a template, in the laboratory and in XBeach.

3 The XBeach model

3.1 Surfbeat mode and governing equations

XBeach (Roelvink et al., 2009) is a depth-averaged (2DH) process-based model designed to simulate storm impacts on beaches, dunes and barrier islands. It was run in *surfbeat* mode, which does not resolve individual short waves but follows their energy envelope on the time scale of wave groups. As a consequence, swash and runup are not resolved wave by wave. This wave energy defines a radiation stress, whose gradients transfer momentum from the short waves to the flow. The flow is solved with the nonlinear shallow-water equations and includes both the mean currents and the infragravity waves generated by the wave groups. It carries sediment in suspension, and sediment flux changes the bed level through the Exner equation:

$$(1 - n_p)\frac{\partial z_b}{\partial t} + \frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} = 0, \tag{1}$$

where n_p is the bed porosity and (q_x, q_y) the sediment flux. The coupling between these modules, and where each varied parameter enters, is summarized in Fig. 1.

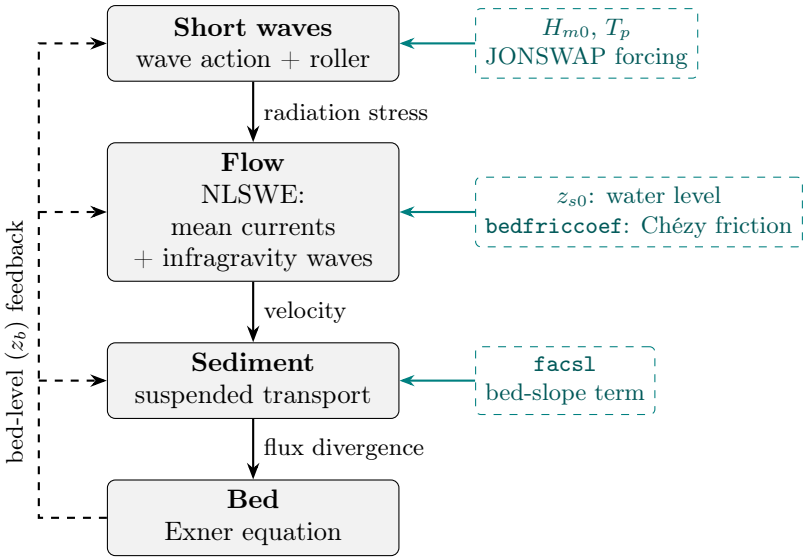


Figure 1: Coupling of the XBeach surfbeat modules. Dashed boxes mark where the parameters varied in the simulation campaign enter the model. The remaining physical parameters are those of Table 1 of Carrion-Bertran et al. (2026), except `random`.

3.2 Storm characterization

The incident waves are prescribed as a JONSWAP spectrum, characterized by the significant wave height H_{m0}, the peak period T_p, the mean wave direction (shore-normal, as in Carrion-Bertran et al. (2026)), the peak enhancement factor ($\gamma = 3.3$) and the directional spreading ($s = 10$), the latter two at their XBeach default values. Carrion-Bertran et al. (2026) forced the model with the time series of a real storm, in which H_{m0}, T_p and the water level vary over the event. Here, instead, the storm is idealized and the wave parameters and the still-water

level z_{s0} are held constant throughout the simulation (reference values $H_{m0} = 2.32\,\mathrm{m}$, $T_p = 12\,\mathrm{s}$, shore-normal incidence and $z_{s0} = 1.042\,\mathrm{m}$). This is less realistic, but it allows the parameters to be varied one at a time.

3.3 Parameters and numerical setup

The setup follows Carrion-Bertran et al. (2026), including their parameters for sediment transport and return flows. The bed-slope coefficient `facsl` defaults to 0.15 in the model build used. The domain covers the P3 profile of Castelldefels (Fig. 2) with 505×600 cells, a uniform alongshore spacing ($\Delta y = 2\,\mathrm{m}$, $L_y = 1202\,\mathrm{m}$) and a cross-shore spacing refined from 10 m offshore to 1 m nearshore. The lateral boundaries are cyclic, so the alongshore Fourier decomposition is exact. Each simulation is a 70 h storm (morphological time; `morfac` = 10, `morfacopt` = 1). All times given in this report are morphological. The wave-group generator is deterministic (`random` = 0), so each combination of wave height and period yields a single wave-group chronology.

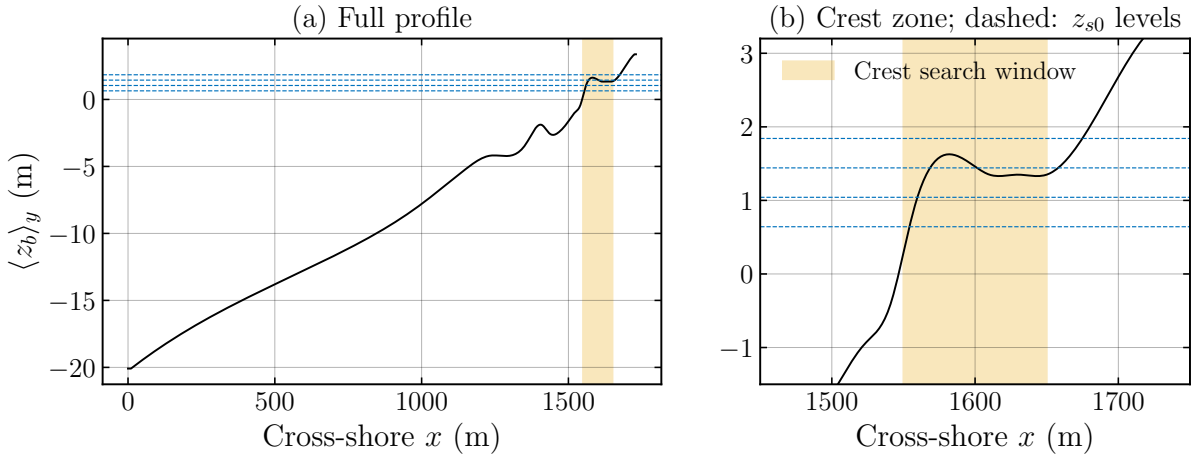


Figure 2: Castelldefels P3 cross-shore profile used in all simulations: (a) full profile; (b) crest zone, with the window in which the crest is searched (shaded) and the still-water levels z_{s0} of the campaign (dashed). For $z_{s0} \leq 1.442\,\mathrm{m}$ the berm crest (1.63 m) can only be overtopped by wave runup; for $z_{s0} = 1.842\,\mathrm{m}$ the still water is above the berm crest, although the higher back-beach topography remains emergent. The storm-impact regimes of Sallenger (2000) compare the extreme water levels, including runup, with the crest of the first line of defence (here the higher back-beach topography), so this last case is not the inundation regime in a strict sense.

4 Experimental design

4.1 Initial perturbation and paired seeds

All simulations start from the alongshore-uniform P3 profile with a small broadband perturbation added to the crest,

$$\eta_c(y) = \sum_j \epsilon \sin\left(\frac{2\pi y}{\lambda_j} + \phi_j\right), \qquad \lambda_j \in [80, 600]\ \mathrm{m}, \tag{2}$$

with random phases ϕ_j and the same amplitude for every wavelength, so that the perturbation does not favour any particular scale (Fig. 3). Each draw of the phases is a *seed*.

Every configuration of physical parameters (a *cell*) was run with the same three seeds (r1, r2, r3), which allows paired comparisons between cells. Fig. 4 compares the bed at the start and at the end of the storm for three cells: the centimetric seed perturbation grows into channels with amplitudes of metres, whose spacing differs between cells even when they start from the same seed.

In addition, the wave-group generator is deterministic (`random` = 0), so comparisons along H_{m0}, z_{s0}, `facsl` and `bedfriccoef` share a single wave-group realization. However, changing T_p produces a different sequence, and part of the variation along T_p may reflect the chronology rather than the parameter. Sampling several chronologies per configuration would be more robust, but it would multiply the cost of a campaign that already comprises 158 runs. Instead, each configuration was run with three seeds of the initial perturbation, which are expected to capture part of this variability.

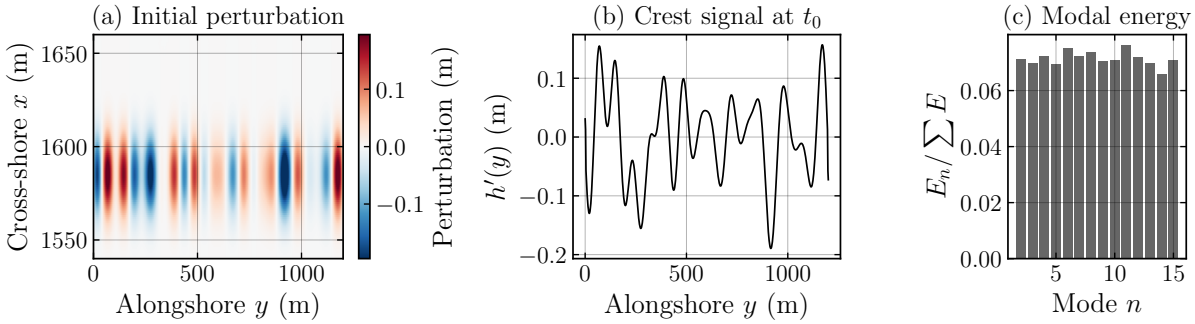


Figure 3: Broadband initial perturbation of seed r3 of the cell $H_{m0} = 2.0\,\mathrm{m}$ (all other parameters at the reference values): (a) perturbation field around the crest; (b) crest signal $h'(y)$ at the first output time; (c) its energy per alongshore mode, nearly uniform across the band 80–600 m.

4.2 Simulation campaign

The campaign comprises 158 runs (Table 1), each sweep varying one or two parameters around a common reference configuration (`facsl` = 0.15, $z_{s0} = 1.042\,\mathrm{m}$, $H_{m0} = 2.32\,\mathrm{m}$, $T_p = 12\,\mathrm{s}$, Chézy 55). Twelve runs stopped before the end of the storm and are excluded. Of the remaining 146, 15 did not develop channels (late spacing below 175 m, close to that of the initial perturbation); these combine a low water level with small waves or a large `facsl`, or have very high friction. The remaining 131 channelized runs are used in two ways. The clock analysis of Section 7 uses them without the 11 runs that repeat, in another block, the same physical configuration and seed (120 runs). The paired statistics of Section 8 use only the 34 cells whose three seeds are complete and channelized (102 runs); the reference configuration appears in two of these cells (M2 and M3).

Sweep	Axes varied	Runs
Water level × wave height	$z_{s0} \times H_{m0}$	36
Bed slope × water level.	`facsl` $\times z_{s0}$	36
Friction and period	`bedfriccoef`, T_p	24
Bed slope at higher friction	`facsl` at `bedfriccoef` = 45	15
Friction at high bed slope	`bedfriccoef` at `facsl` = 5	15
Wave height (Hm0)	H_{m0}	12
Friction and water level	`bedfriccoef`, z_{s0}	12
Bed slope, extended	`facsl`	8
Total	48 physical configurations	158

Table 1: Simulation campaign: sweeps, parameters varied and number of runs.

5 Scale estimators

Counting channels by eye is not reliable. In the same cell, some seeds seem to have three channels and others four or five, and the deposits between channels have unequal sizes (Fig. 5). An

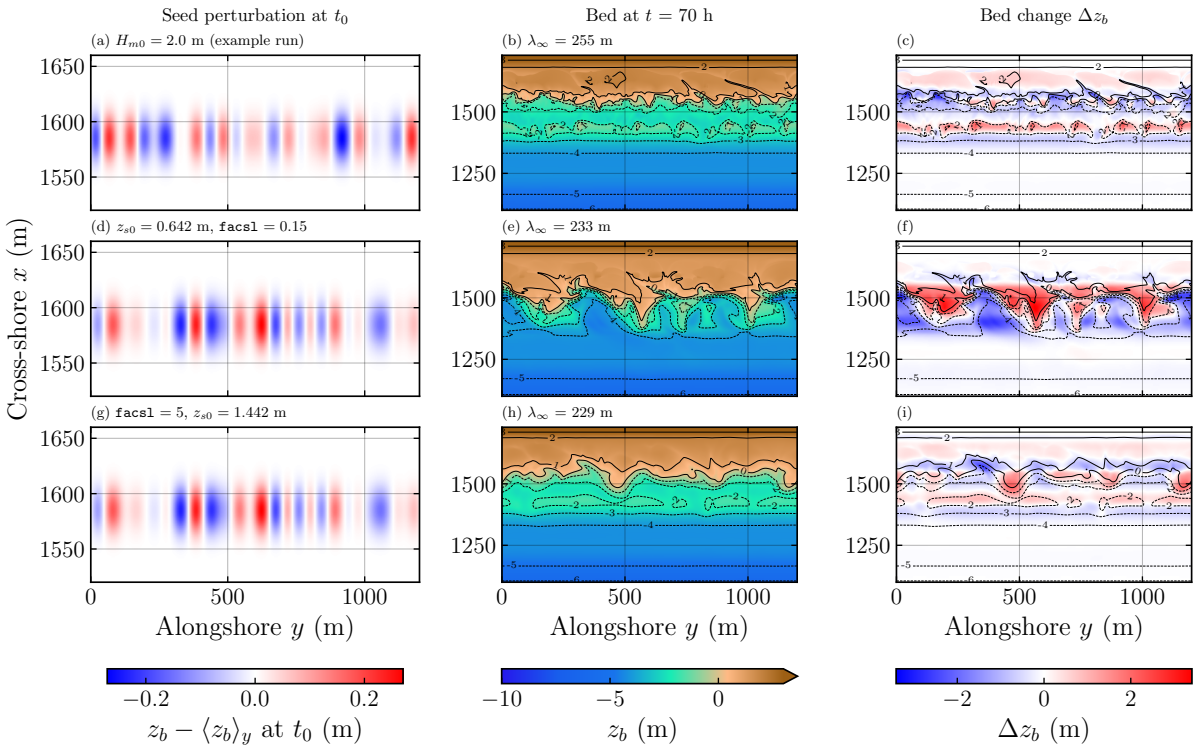


Figure 4: Seed perturbation (left), bed at the end of the storm (centre) and bed change Δz_b between the start and the end (right) for three cells: (a–c) the example run `Hm020_r3`; (d–f) $z_{s0} = 0.642$ m and (g–i) `facsl` $= 5$, $z_{s0} = 1.442$ m, both with seed r1, so they start from the same bed. The left column is zoomed on the crest, where the perturbation is added. Each column has a common colour scale. Contours: z_b every 1 m (dashed below 0). λ_∞ is the Fourier centroid averaged over 30–70 h.

objective, automatic estimator of the spacing is needed. This section compares a valley-detection estimator, the Fourier centroid of a crest signal and a wavelet-based estimator. Unless stated otherwise, the single-run figures of this section use the *example run* `Hm020_r3` ($H_{m0} = 2.0\,\mathrm{m}$, all other parameters at the reference values, seed r3).

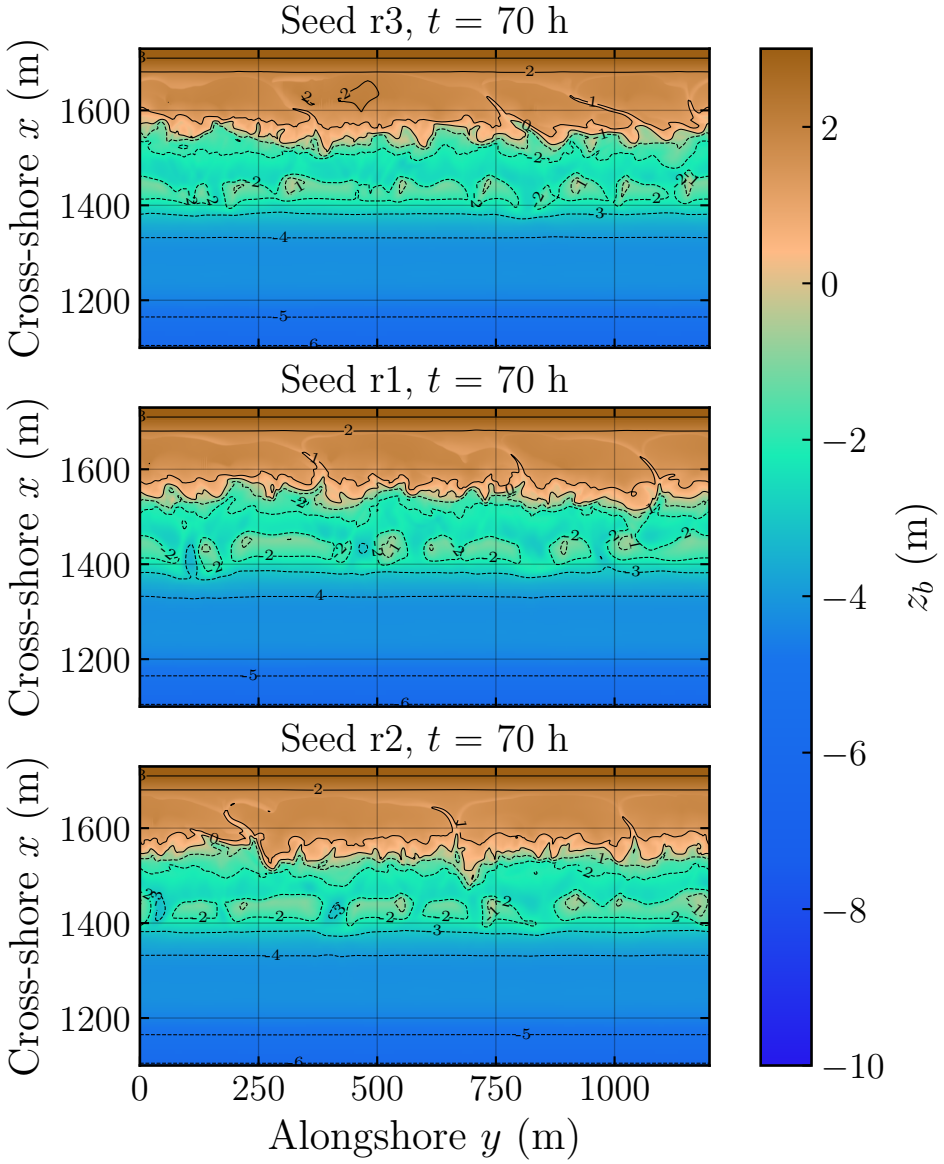


Figure 5: Final bed level of the three seeds of the cell $H_{m0} = 2.0\,\mathrm{m}$ (other parameters at the reference values): the number of channels is ambiguous by eye. Contours every 1 m (dashed below 0).

5.1 Valley detection

Carrion-Bertran et al. (2026) measured the spacing as the mean distance between channels, detected as valleys of the alongshore bed profile taken where the deposition is largest. Valleys had to lie below a bed-level threshold derived from the variability in the depositional zone, and a minimum separation between them was imposed. Fig. S1 illustrates a simple implementation on the final crest signal of one run (valleys with a prominence of at least half the standard deviation of the signal and at least 40 m apart). It finds five valleys, i.e. $L_y/5 = 240\,\mathrm{m}$, close to the Fourier centroid of the same signal, 242 m. Even when the two agree, a count of channels in a periodic

domain can only take the values L_y/N (300, 240 and 200 m for $N = 4, 5, 6$), and whether a shallow dip counts as a channel depends directly on the thresholds. These two limitations are the main reasons for adopting a spectral estimator. The implementation shown here is illustrative and does not reproduce the exact procedure of Carrion-Bertran et al. (2026).

5.2 Crest extraction

The 2D morphology is reduced to a 1D alongshore crest signal suitable for spectral analysis. At each output time, the alongshore mean of the bed level is removed and the remaining perturbation is averaged in the cross-shore direction with a Gaussian weight of width $L_x = 20$ m centred on the crest position $x_c(t)$:

$$h'(y,t) = \sum_x w(x;t)\,\big[z_b(x,y,t) - \langle z_b\rangle_y(x,t)\big], \qquad w(x;t) \propto \exp\left[-\left(\frac{x - x_c(t)}{L_x}\right)^2\right], \tag{3}$$

with $\sum_x w = 1$ (Fig. S2). The window must follow the crest, which migrates landward during the storm. In run `Hm020_r1` (Fig. S3), whose crest migrates 47 m, a window fixed at the initial crest position ends up 21 m seaward of it at 20 h. The crest is tracked with hysteresis to avoid jumps between maxima with similar heights once the crest erodes during the storm, and the tracked position is then smoothed out over these positions.

5.3 Fourier centroid

With cyclic lateral boundaries, the alongshore signal $h'(y,t)$ can be written exactly as a Fourier series,

$$h'(y,t) = \sum_n \hat{h}'_n(t)\, e^{2\pi i n y/L_y}, \tag{4}$$

where the integer n is the number of wavelengths that fit in the domain, so that mode n has wavelength $\lambda_n = L_y/n$, and $\hat{h}'_n(t)$ is its complex amplitude, computed with the discrete Fourier transform of h' over the $N = 601$ alongshore grid points.

The energy of mode n is $E_n(t) = |\hat{h}'_n(t)|^2$, and the active band is $n = 2, \ldots, 15$, which for $L_y = 1202$ m corresponds to wavelengths between 80 and 600 m. The energy is distributed over several modes, and the most energetic one changes in time. More importantly, it is not reproducible: seeds of the same cell end with different dominant modes, while their energy falls in a common band (Fig. 6). For this reason we measure a characteristic scale instead of the dominant mode, using the spectral centroid

$$n_F(t) = \frac{\sum_{n\in B} n\,E_n(t)}{\sum_{n\in B} E_n(t)}, \qquad \lambda_F(t) = \frac{L_y}{n_F(t)}. \tag{5}$$

The spacing λ_∞ is the mean of λ_F over 30–70 h, since at the beginning the centroid is always fixed from the seed. This estimator proved to be robust and consistent with qualitative observations of the profiles, although it is biased by the higher modes and underestimates scale.

5.4 Continuous wavelet transform

Following Lazarus and Armstrong (2015), the spacing was also estimated with the continuous wavelet transform (CWT). The Fourier spectrum is global: it gives the energy of each scale, but not where along the coast that scale is present. The CWT resolves both by convolving the signal with a short oscillating pattern, the *wavelet*, stretched to different sizes. We use the Morlet wavelet (Torrence and Compo, 1998), a complex sinusoid modulated by a Gaussian envelope,

$$\psi(\eta) = \pi^{-1/4}\, e^{i\omega_0\eta}\, e^{-\eta^2/2}, \qquad \omega_0 = 6. \tag{6}$$

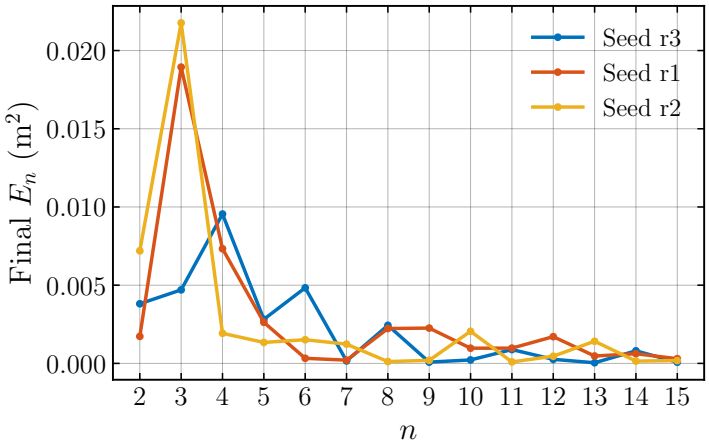


Figure 6: Final alongshore spectra per seed for the cell $H_{m0} = 2.0\,\mathrm{m}$ (other parameters at the reference values): the dominant mode changes between seeds, but the energy falls in a common band.

Stretching it by a scale s and normalizing gives $\psi_s(y) = s^{-1/2}\,\psi(y/s)$, and the transform of h' at scale s is the convolution

$$W(y, s, t) = (h' * \psi_s)(y, t) = \int_0^{L_y} h'(y', t)\,\psi_s(y - y')\,dy', \tag{7}$$

with h' extended periodically, consistent with the cyclic domain. In practice it is computed in Fourier space, where the convolution becomes a product. W is large where the signal locally resembles an oscillation of size s, and $|W|^2$ measures how much of the signal at y is carried by that scale.

5.5 Wavelet estimator

The estimator follows the idea of Lazarus and Armstrong (2015), who averaged the wavelet power alongshore to obtain a spectrum and read the dominant spacing from it. Ours is built in three steps.

1. *Local power.* Before the transform, the Fourier modes with $\lambda > 600\,\mathrm{m}$ are removed from h'; for $L_y = 1202\,\mathrm{m}$ these are $n = 1$ and $n = 2$, long undulations of the crest that are not channels. The power is then divided by the scale, $P = |W|^2/s$ (Liu et al., 2007): with this rectification, sinusoids of equal amplitude give equal power at all scales, whereas $|W|^2$ alone grows with s and biases the estimate towards large scales.

2. *Alongshore average.* Averaging P along the coast gives a single spectrum,

$$\overline{P}(s, t) = \frac{1}{L_y} \int_0^{L_y} P(y, s, t)\,dy, \tag{8}$$

   which says how much of the crest signal is carried by each scale (Fig. 7).

3. *Centroid.* The spacing is the power-weighted mean wavelength of that spectrum,

$$\lambda_W(t) = \frac{\displaystyle\int_{s_{\min}}^{s_{\max}} \overline{P}(s, t)\,\lambda_s\,\frac{ds}{s}}{\displaystyle\int_{s_{\min}}^{s_{\max}} \overline{P}(s, t)\,\frac{ds}{s}}, \tag{9}$$

   where the measure ds/s gives equal weight to equal ratios of scale, $\lambda_s = a\,s$ is the Fourier wavelength of scale s, with $a = 4\pi/(\omega_0 + \sqrt{2 + \omega_0^2}) \approx 1.03$ for the Morlet wavelet (Torrence and Compo, 1998), and the scales span a logarithmic grid of 80–800 m.

The centroid, rather than the maximum of the spectrum, is essential in a periodic domain. There h' has energy at the wavelengths close to L_y/n, and the maximum falls on one of them. It locks to the domain modes and jumps between them when two modes have similar energy (Fig. 8). The centroid instead varies continuously with the distribution of energy, in the same way as the Fourier centroid λ_F.

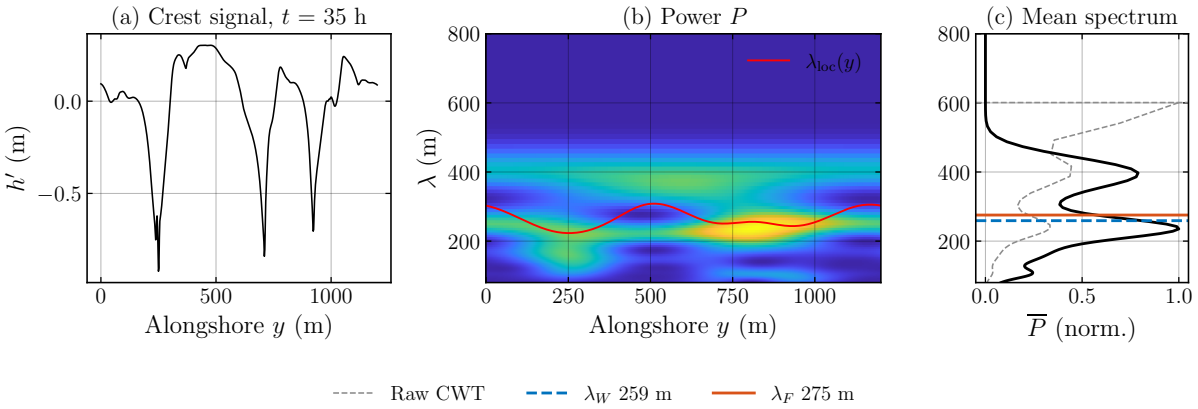


Figure 7: Continuous wavelet transform of the crest signal of the example run `Hm020_r3` at 35 h: (a) crest signal; (b) power P used by the estimator (modes with $\lambda > 600$ m removed, $P = |W|^2/s$), with the local centroid in scale (red line); (c) alongshore-averaged power $\overline{P}$, normalized, with the resulting estimate and the Fourier centroid λ_F: $\lambda_W = 259$ m and $\lambda_F = 275$ m. Dashed grey: the same average for the raw CWT (unrectified $|W|^2$ on a grid in $n = 2$–15) and its maximum, as in Fig. 8. The full version, also at the end of the storm, is Fig. S4.

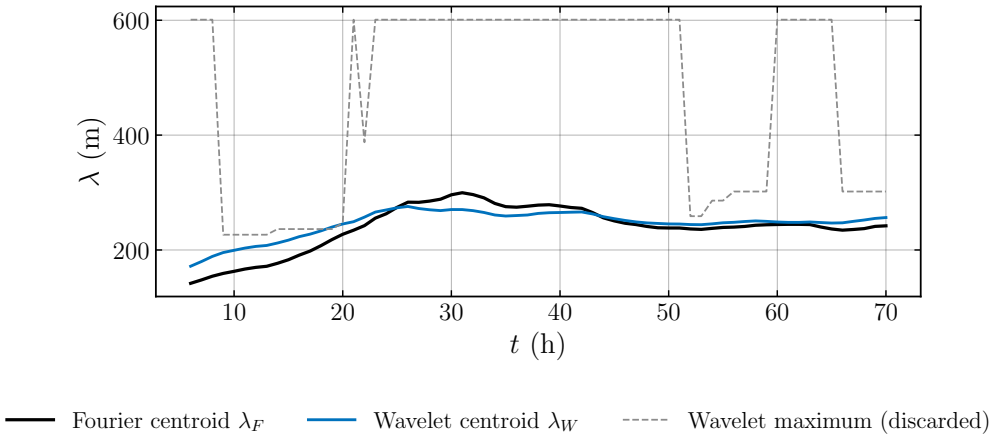


Figure 8: Spacing of the example run `Hm020_r3` with the Fourier centroid λ_F (black), the wavelet centroid λ_W (blue) and the maximum of the wavelet spectrum (grey, discarded), which locks to the domain modes.

5.6 Domain size

The estimators are first tested on the domain size, which also verifies that the spacing is not imposed by the size of the periodic domain. If the selected spacing were an artefact of the modes allowed by the domain, it would change with L_y; if it is physical, n_F should scale with L_y and λ_F should remain approximately constant.

This was tested with four seeds in each of three domains, $L_y = 1000$, 1600 and 2400 m, and with one run in a domain of $L_y = 4096$ m, keeping the band fixed in wavelength (80–600 m), i.e.

with the range of n scaled with L_y.

The curves of all domains overlap until about 25 h (Fig. 9). Afterwards, with λ_F the mean over 30–70 h is 285, 281 and 246 m for $L_y = 1000$, 1600 and 2400 m, and 231 m in the single run with $L_y = 4096$ m. With λ_W they are 345, 340, 290 and 274 m. The seeds of the different domains overlap (Fig. 9c), and the differences between domains (14 % with λ_F and 16 % with λ_W between 1000 and 2400 m) are small compared with the change of a factor 2.4 in L_y that an artefact of the domain would produce. The domain size therefore does not appear to fix the spacing, although a weak effect cannot be excluded with four seeds per domain.

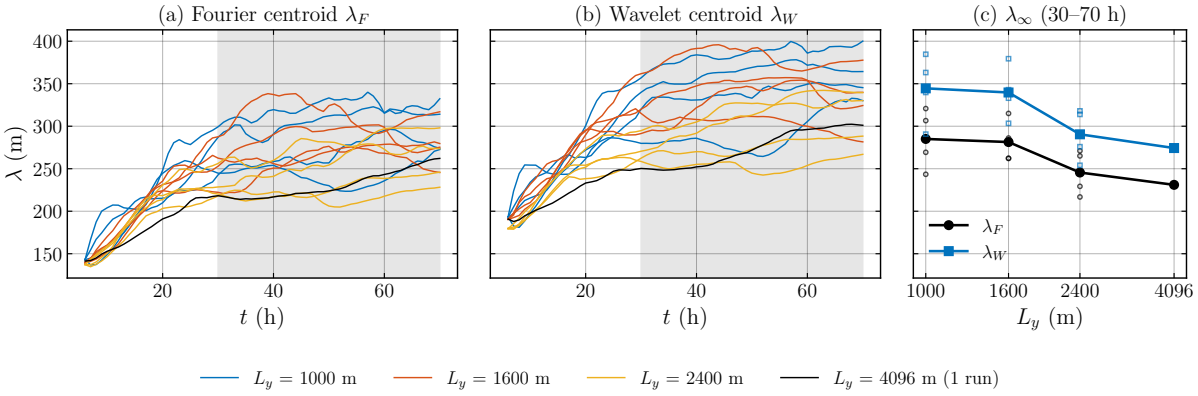


Figure 9: Domain-size test: (a) Fourier centroid λ_F and (b) wavelet centroid λ_W vs. time for $L_y \in \{1000, 1600, 2400\}$ m, four seeds each, and one run with $L_y = 4096$ m (black, first 70 h of a 250 h storm), with the same wavelength band (80–600 m), shaded, the 30–70 h window. (c) Late spacing λ_∞ (mean over 30–70 h) versus L_y with both estimators: seeds (small open symbols) and means. The seeds of the different domains overlap.

6 Smoothing and incising processes

Channels form where the overwash flow incises the deposits, as described by Carrion-Bertran et al. (2026): the overtopping flow decelerates and deposits over incipient deposits, while the return flow accelerates between them and cuts channels, but incision alone does not fix a spacing. The numerical model of Lazarus and Armstrong (2015) combines competition between throats for a limited discharge with an alongshore diffusion of the barrier. Among all its parameters, the diffusivity is the one that most controls the wavelength, with higher diffusivity giving longer wavelengths, although they note that diffusive smoothing is not essential to the pattern-forming dynamics, which they attribute to the competition between throats. This suggests, as a working hypothesis, that the spacing is set not by the flow alone but by the competition between a process that incises the deposits cross-shore and one that smooths them alongshore.

This competition is formalized in the model of Perron et al. (2009) for evenly spaced ridges and valleys:

$$\frac{\partial z}{\partial t} = D\,\nabla^2 z - K A^m\,|\nabla z| + U, \tag{10}$$

with D a diffusivity, A the drainage area, K and m constants of the incision law and U the uplift rate. Diffusion damps short wavelengths and incision amplifies valleys, and the two balance at the characteristic length

$$L_c = \left(\frac{D}{K}\right)^{1/(2m+1)}, \tag{11}$$

to which the valley spacing is taken to be proportional (Perron et al., 2009), independently of the uplift rate. Although developed for landscapes, it offers a quantitative test for XBeach if D and K can be identified. The bed-slope term of XBeach, scaled by `facsl`, can be approximated

by a bed diffusion in the Exner equation (Sect. S2.1) and is the natural candidate for D. With $D \propto$ `facsl` and K set by the hydrodynamics, the spacing should increase with `facsl` with slope $+1/3$ ($m = 1$). This identification is an assumption. If the test fails, another process could still act as an effective D.

The campaign varies the parameters that act on each side of the balance. `facsl` acts on the smoothing, and the other parameters act on the incision. z_{s0} and H_{m0} control the volume of water that overtops the crest, for which the throats compete. T_p controls the wave groups and the infragravity motion that drive overwash, and the bed friction controls the velocity, and hence the incising capacity, of the overwash and return flows. Sweeping `facsl` at two friction levels tests whether smoothing and incision act through their ratio, as D/K implies. The predictions of each mechanism are summarized in Table 2.

Table 2: Candidate mechanisms for the selection of the spacing, their control variable and their prediction.

Mechanism	Control variable	Prediction	Test (Sect. 7, 8)
Diffusion–incision (Perron et al., 2009)	D/K, with $D \propto$ `facsl`	λ increases with `facsl`; slope $+1/3$ ($m = 1$)	`facsl` sweeps at two frictions
Flow competition (Lazarus and Armstrong, 2015)	Overwash volume	The parameters that control the overwash volume should affect λ	Clock V_{ov}
Hydrodynamic feedback (Carrion-Bertran et al., 2026)	Flow structure over the deposits	λ follows the hydrodynamic parameters, not `facsl` alone	H_{m0}, T_p, z_{s0}, friction; clock Σ^+

6.1 Flow competition and hydrodynamic feedback

The other two candidates give no closed law for the spacing and are used as qualitative guides. In the model of Lazarus and Armstrong (2015), the throats compete for a limited overwash discharge. If this competition selects the spacing, the parameters should act on λ through the volume of water that overtops the crest, and comparing runs at equal overwash volume V_{ov} should remove most of their effect. In the feedback of Carrion-Bertran et al. (2026), the overtopping flow decelerates over incipient deposits and the return flow accelerates between them. The spacing should then follow the hydrodynamic parameters (H_{m0}, T_p, z_{s0} and friction) rather than `facsl`, possibly through the amount of sediment the flow deposits, measured by the deposited volume Σ^+.

7 Morphological clocks

The spacing coarsens through the storm by discrete fusions, so comparing configurations at the same storm time mixes two possible effects of a parameter: on the scale that is finally selected, and on how fast the storm drives the coarsening. The configurations are therefore also compared at an equal value of a cumulative *clock*, a variable that measures how far the storm has progressed. If a clock accounts for the effect of a parameter, the spacing at equal clock should not depend on that parameter. The overwash volume checks to what extent the parameters act through the volume of water that overtops the crest, as the model of flow competition suggests, and the deposited volume is its morphological counterpart. The comparison is made along the one-factor-at-a-time (OFAT) axes of the campaign, each varying one parameter around the reference configuration.

7.1 Clocks and procedure

Three clocks are compared, all cumulative from the start of the storm (details in Sect. S3.1):

- the storm time t, in hours of morphological time;
- the overwash volume $V_{ov} = \int Q_{ov}\,dt$, where Q_{ov} is the landward discharge through the section of minimum water depth near the crest;
- the deposited volume Σ^+, the cumulative sum of the positive bed-level changes between consecutive outputs in the band $1450 \leq x \leq 1750$ m.

The incident wave energy $\int H_{m0}^2\,dt$ is proportional to t except along H_{m0}, where it does not reduce the spread (Sect. S3.2). The storm time serves as the reference. All the runs of an axis start from the same initial perturbations, so their spacings are initially equal and their differences grow during the storm. Reading them midway through this growth therefore reduces their differences for any clock. A clock is informative only if it reduces the spread more than the storm time itself, read at the same point of the growth.

For each OFAT axis and each clock, λ (smoothed over 5 h) is compared with the late spacing λ_∞ of the same run (30–70 h; 50–70 h as control). Each run is read when its clock reaches a reference value c^*, the first value at which the median run of the axis has covered 60 % of the way from its initial spacing to λ_∞ (Fig. S7). Equal time corresponds to 16–21 h.

The *residual spread* is the fraction of the final difference between levels that remains at equal clock (0: the clock accounts for the whole effect of the parameter; 1: for none). It is computed as the range of $\log\lambda$ between the levels of the axis at c^* divided by the range of $\log\lambda_\infty$, and it is larger than 1 if the levels differ more at c^* than in λ_∞. Levels that do not reach c^* are left out. The test uses the 120 channelized runs without repetitions, so the axes have more levels than in Table 3.

7.2 Results

The spacing at equal clock is compared with λ_∞ for the H_{m0} and T_p axes in Fig. 10, and the residual spread along every axis is summarized in Fig. 11 (values in Table S1; all axes in Figs. S8 and S9). The residual spread is computed for each seed; the values below are means over the seeds, for λ_F / λ_W with the 30–70 h reference. A clock is counted as better than time only if this holds for every seed.

Along H_{m0} the residual spread is already small at equal time (0.25 / 0.22), and that at equal Σ^+ (0.22 / 0.17) is not distinguishable from it. Along T_p, at equal Σ^+ the mean residual spread is about half of that at equal time, with both estimators (0.48 against 0.90 with λ_F, 0.42 against 0.70 with λ_W) and also in the control window (Table S1). Seed by seed, this holds only with λ_W and the control window (0.37 ± 0.17 against 0.61 ± 0.07, where the error is the largest deviation of a seed from the mean). Along the friction axis Σ^+ is better than time for every seed with λ_F (0.69 ± 0.15 against 1.01 ± 0.12), but not with λ_W. The overwash volume does not reduce the spread of any axis more than time, and along all axes except T_p some levels do not reach its reference value.

The value $c_{1/2}$ of each clock at which λ has completed half of its growth varies between runs by a factor of 1.38 in t and 3.39 in V_{ov} (1.41 and 3.56 with λ_W; Fig. S10). In these simulations the coarsening appears to be paced more by the storm time than by the volume of water that has overtopped the crest.

Two caveats apply. The deposited volume is an output of the same simulations, so a small residual spread at equal Σ^+ indicates that the spacing and the deposition evolve together, not that one causes the other. In addition, each value of T_p comes with its own wave-group chronology, and the residual spreads depend on the choice of c^*.

8 Experiments

8.1 Statistical method

Each run is reduced once to time series and scalars (spacing, overwash discharge and volume, deposited volume, crest relief), which are collected in a table with one row per run; all statistics

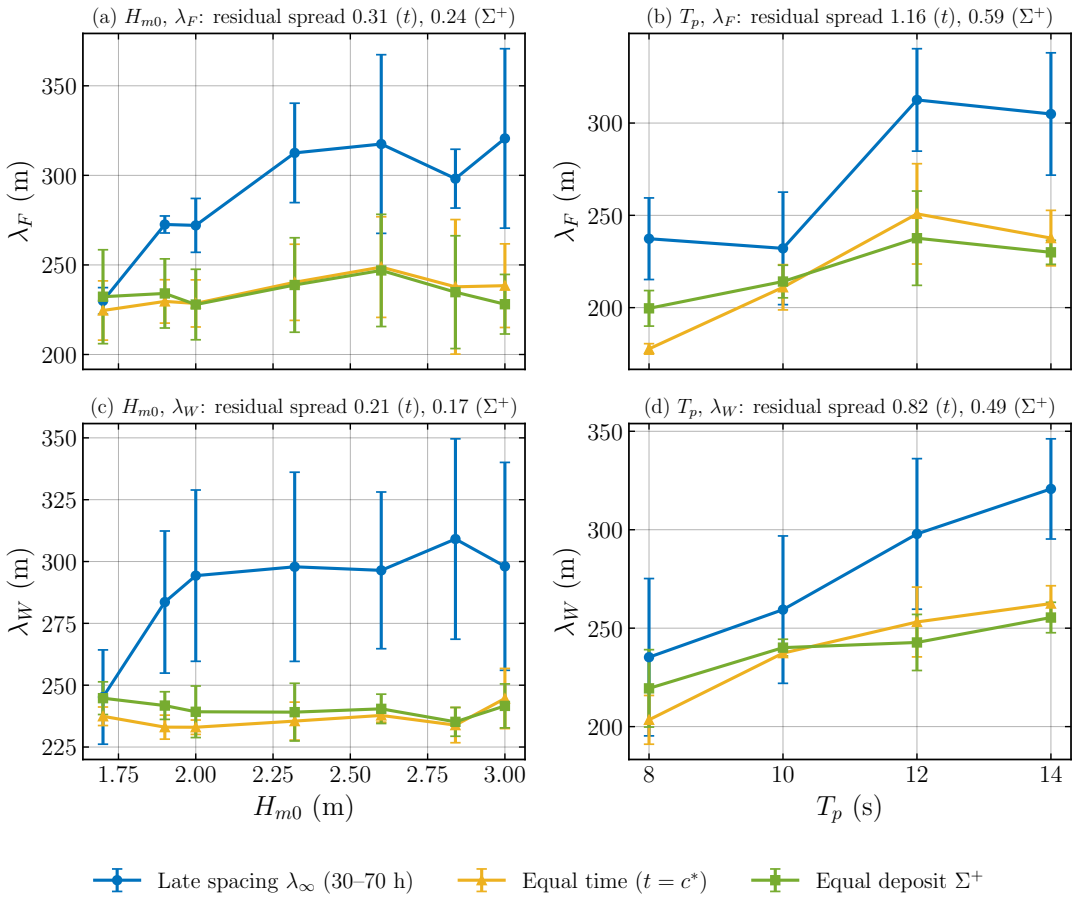


Figure 10: Equal-clock test on two OFAT axes, with λ_F (top) and λ_W (bottom): late spacing λ_∞ (30–70 h) and spacing at equal time $t = c^*$ and at equal deposited volume $\Sigma^+ = c^*$ (mean ± s.d. over the runs of each level). (a, c) H_{m0}; (b, d) T_p. Panel titles: residual spread of the seed-averaged spacing; seed by seed in Table S1.

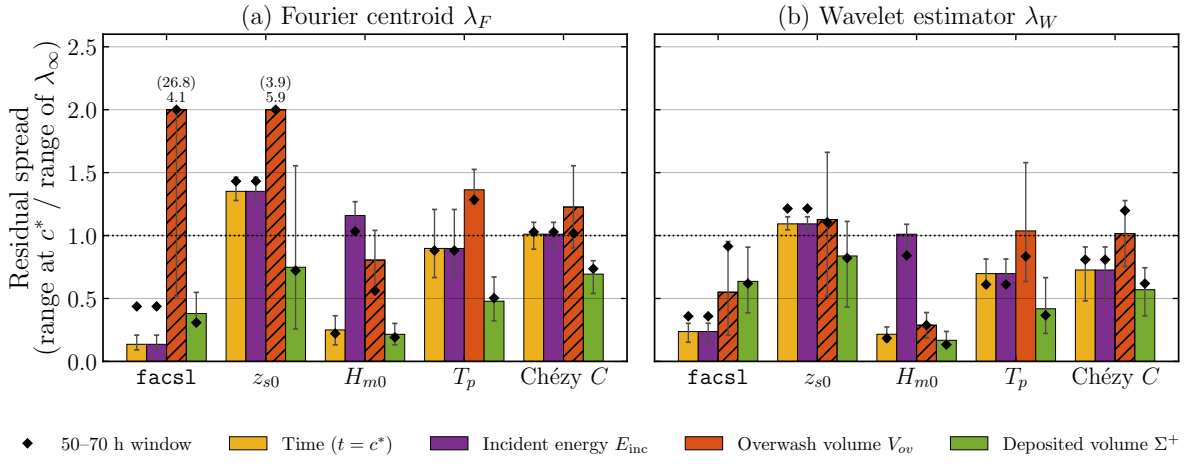


Figure 11: Residual spread of each OFAT axis at equal clock, with (a) λ_F and (b) λ_W: bars, mean over the three seeds (30–70 h), with the seed-to-seed range as whiskers; diamonds, control window 50–70 h (values in Table S1). 0: the clock accounts for the whole effect of the parameter; dotted line (1): for none; hatched: some levels do not reach c^*; values above 2 are drawn at 2 and labelled. A clock beats t only if its whisker lies entirely below that of t.

are computed on this table. Unless stated otherwise, λ_∞ is the mean of $\lambda(t)$ over 30–70 h, and the control window 50–70 h is reported alongside.

Sensitivities are estimated along a one-factor-at-a-time (OFAT) star around the reference configuration. For each seed, $\log \lambda_\infty = a + b \log X$ is fitted along the axis X, and the slope b is the mean over the three seeds, with a Student-t interval with 2 degrees of freedom (factor 4.303 at 95 %). These are partial sensitivities, the other parameters are held at their reference values. Fits over the whole set would mix the axes of an unbalanced design, so they are not used to discuss mechanisms. Every result is given for both λ_F and λ_W.

8.2 Seed variability and coarsening

In $\log \lambda_\infty$, the standard deviation between the seeds of a cell is $\sigma_{\text{seed}} = 0.111$, the uncertainty of a cell mean over three seeds is $\sigma_{\text{mean}} = 0.064$, and the dispersion between cells is 0.165, i.e. 2.6 times σ_{mean} (Fig. S11). With λ_W the seed noise is similar (0.109 and 0.063), but the dispersion between cells is smaller, 0.111, i.e. 1.8 times σ_{mean}. The parameters thus appear to move the spacing above the seed noise, more clearly with λ_F. The dominant mode changes in discrete steps, while λ_F varies continuously because it averages over the modes (Fig. 12).

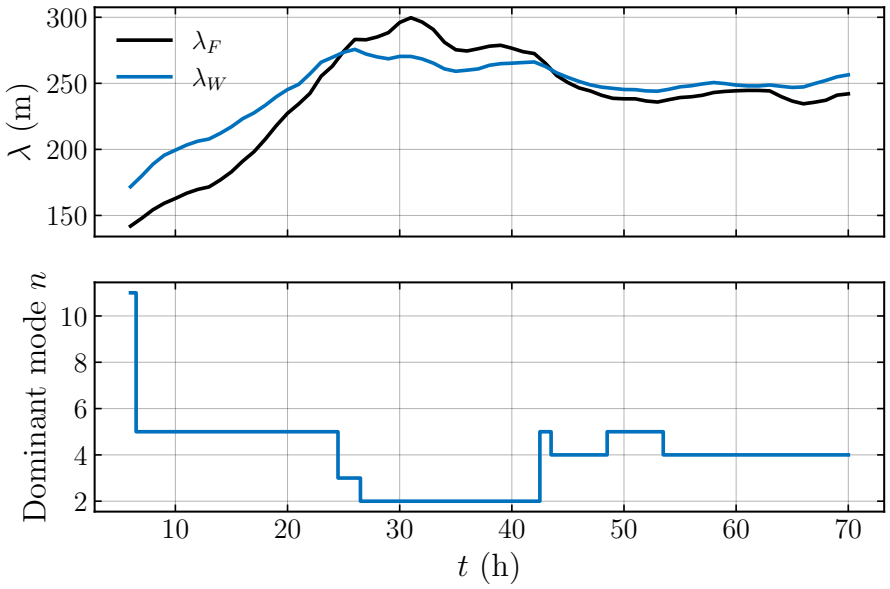


Figure 12: Spacing $\lambda_F(t)$ and $\lambda_W(t)$ (top) and dominant mode (bottom) of the example run `Hm020_r3`. Between about 27 and 43 h the dominant mode is $n = 2$, which λ_W excludes.

8.3 Partial sensitivities

The OFAT slopes are given in Table 3 and the responses along each axis in Fig. 13. With λ_F, the slopes of T_p, H_{m0} and z_{s0} are between +0.4 and +0.6 and their intervals exclude zero, while `facsl` has a small negative slope, -0.08 ± 0.03. The friction response is not monotonic, with a maximum near Chézy 70 with both estimators, and the interval of its slope is wide.

With λ_W the point estimate for T_p is similar (+0.6), those for z_{s0} and H_{m0} are about half as large (+0.2 and +0.3), and that for `facsl` is −0.03. All intervals are wider and include zero. The cell means of the two estimators are correlated ($r = 0.85$ in $\log \lambda_\infty$; Fig. S12).

8.4 Diffusion–incision test

Along `facsl` the measured slope is -0.08 ± 0.03 with λ_F and -0.03 ± 0.04 with λ_W, over almost two decades of `facsl`. The upper bounds of the intervals remain well below the predicted +1/3 (Sect. 6) with both estimators and both windows (Table 3, Fig. 14). The same sweep at Chézy 45 gives slopes compatible with zero (Sect. S2.3).

	Fourier centroid λ_F		Wavelet estimator λ_W	
Axis	30–70 h	50–70 h	30–70 h	50–70 h
T_p	$+0.60 \pm 0.17$	$+0.50 \pm 0.09$	$+0.6 \pm 1.0$	$+0.6 \pm 0.9$
H_{m0}	$+0.6 \pm 0.3$	$+0.7 \pm 0.6$	$+0.3 \pm 0.8$	$+0.3 \pm 0.8$
z_{s0}	$+0.4 \pm 0.3$	$+0.37 \pm 0.13$	$+0.2 \pm 0.6$	$+0.2 \pm 0.6$
`bedfriccoef`	$+0.1 \pm 0.9$	0.0 ± 1.0	-0.2 ± 0.9	-0.1 ± 1.1
`facsl`	-0.08 ± 0.03	-0.077 ± 0.014	-0.03 ± 0.04	-0.02 ± 0.06

Table 3: One-factor-at-a-time sensitivities: slope b of $\log \lambda_\infty = a + b \log X$, given as mean $\pm$ half-width of the 95 % Student-t interval across the three seeds (2 d.o.f.), for the main (30–70 h) and control (50–70 h) windows and the two estimators. Levels: T_p 8, 10, 12, 14 s; H_{m0} 1.7–3.0 m (5 levels); z_{s0} 0.642–1.442 m (5); `bedfriccoef` 45–90 (5); `facsl` 0.15–8 (6). The Chézy 30 cell is not included because not all of its seeds channelized.

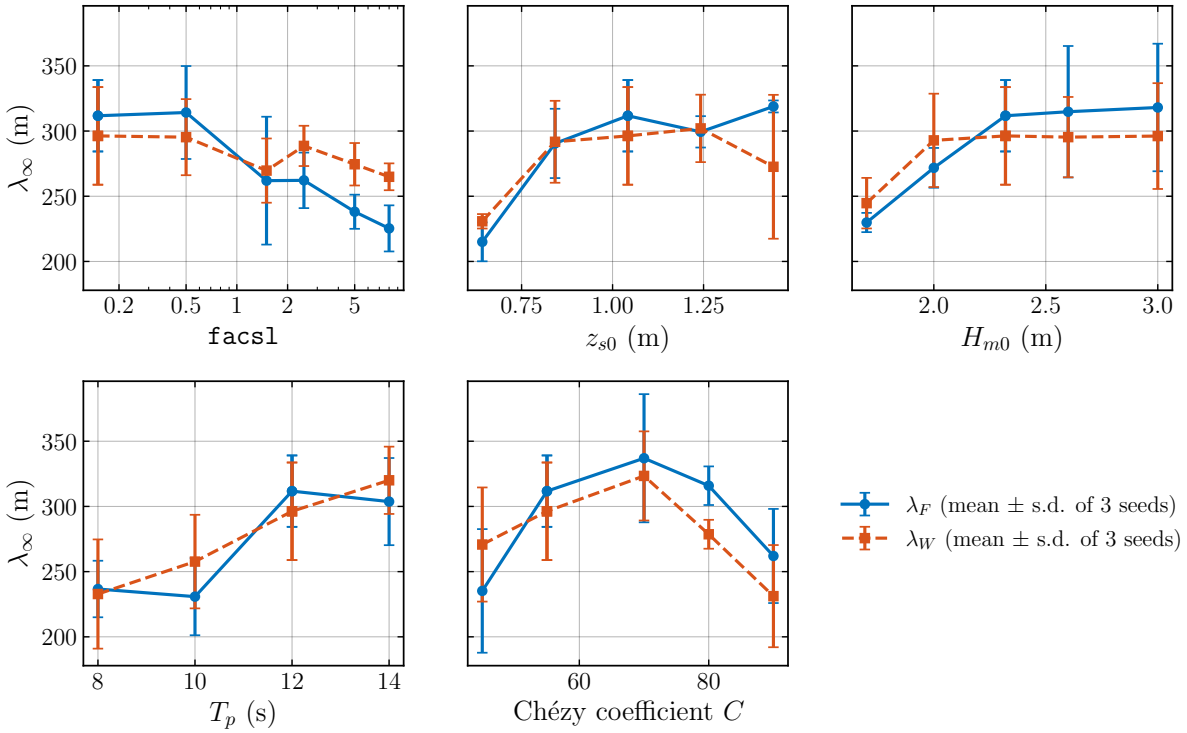


Figure 13: Late spacing λ_∞ (30–70 h) along each axis of the OFAT star, with λ_F (blue, solid) and λ_W (orange, dashed); mean $\pm$ s.d. of the three seeds of each cell (only cells with three complete, channelized seeds).

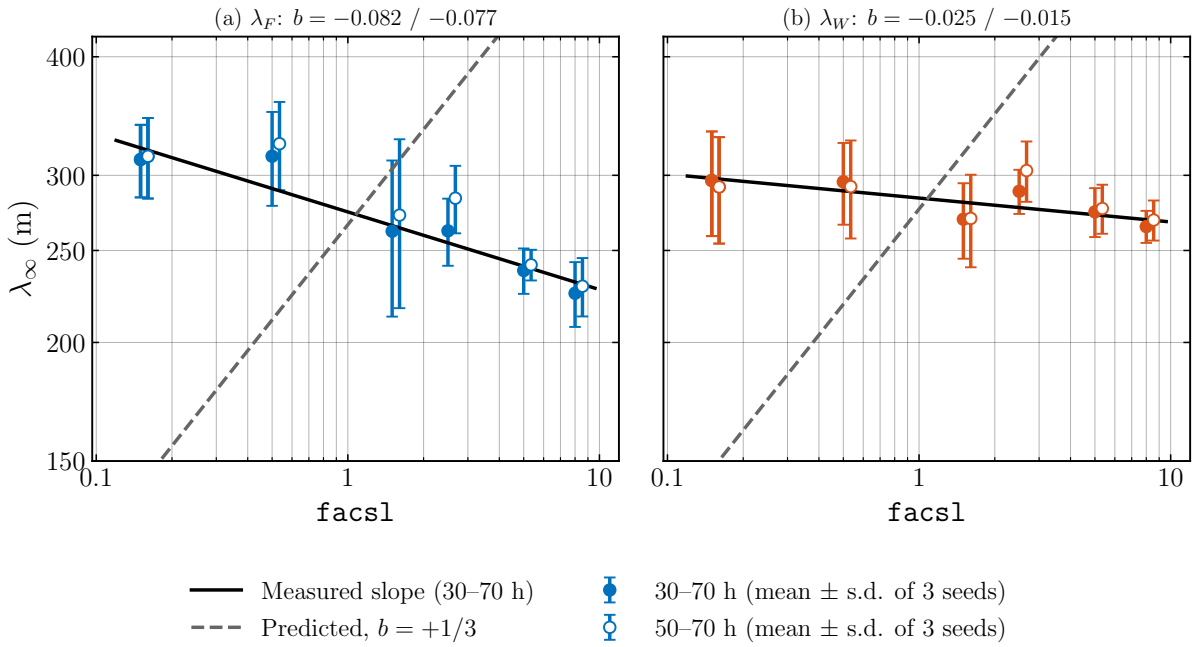


Figure 14: Diffusion–incision test: λ_∞ versus `facsl` with (a) λ_F and (b) λ_W (mean ± s.d. of three seeds; filled 30–70 h, open 50–70 h). Solid: fitted slope; dashed: predicted +1/3, through the centroid of the points.

8.5 Flow-competition test

As argued in Sect. 6.1, if the throats compete for the overwash discharge, the spacing should collapse at similar V_{ov}. It does not, with both estimators and both windows, V_{ov} does not reduce the spread of any OFAT axis more than t (Sect. 7.2). Pairs of cells with equal V_{ov} and incompatible spacings appear with λ_F, but rarely with λ_W (Sect. S5.1). Note that V_{ov} is computed from hourly output and is therefore approximate (Sect. S5.2).

9 Discussion

We now discuss the three candidate mechanisms in turn. We say that a parameter modulates the spacing if it moves it up or down, and that a process selects it if it sets its value.. A statement is retained only if it holds with both estimators (Table 4).

9.1 Diffusion–incision

Under the mapping $D \propto$ `facsl`, the balance of Perron et al. (2009) predicts that the spacing grows with `facsl` with a slope of +1/3. The measured slope is negative with λ_F and close to zero for λ_W.Its upper bound stays well below +1/3 with both estimators and both windows (Table 3). The results therefore do not support `facsl` acting as the diffusivity that selects the spacing. `facsl` does affect the outcome: at large values, together with low water levels or small waves, it is associated with runs that do not develop channels. Its role appears closer to damping the instability than to fixing its scale. The test is conditional on the assumed mapping $D \propto$ `facsl`.Another process, such as avalanching (`wetslp`), could act as an effective diffusivity and was not varied.

9.2 Flow competition

V_{ov} does not account for the effect of any parameter better than the storm time (Sect. 8.5), and the pace of the coarsening varies more between runs in V_{ov} than in t (Sect. 7.2). The overwash

volume may therefore modulate the spacing, through z_{s0} and H_{m0}, but it does not appear to select it.

Note that V_{ov} is approximate (hourly output). Also, in XBeach the discharge over the crest is controlled by wave groups and infragravity motion, so a closure in volume alone may not translate the mechanism proposed by Lazarus and Armstrong (2015) from their laboratory experiment and cellular model.

9.3 Hydrodynamic feedback

T_p, H_{m0} and z_{s0} all move the spacing upwards, with intervals that exclude zero with λ_F (Table 3). This is consistent with the feedback of Carrion-Bertran et al. (2026), in which the spacing follows the flow over the deposits. Along T_p the effect is reduced more at equal Σ^+ than at equal time on average, but seed by seed only with λ_W and the control window; along the friction axis Σ^+ is better than time for every seed only with λ_F; along H_{m0} the residual spread is already small at equal time. These parameters may act on the spacing partly through how much sediment the storm deposits, but this result is not robust.

With λ_W the slopes of H_{m0} and z_{s0} are about half as large and all intervals include zero. The issues of Sect. 7.2 on Σ^+ and the wave-group cronology also apply here. The hydrodynamic feedback is the candidate most consistent with the results, but it gives no quantitative law for λ_∞.

9.4 Robustness

	λ_F		λ_W	
Statement	30–70 h	50–70 h	30–70 h	50–70 h
`facsl` slope below +1/3 (Table 3)	yes	yes	yes	yes
T_p, H_{m0}, z_{s0} slopes positive (intervals exclude 0)	yes	yes	no	no
V_{ov} reduces the spread of no axis more than t (Table S1)	yes	yes	yes	yes
Σ^+ reduces the spread along T_p more than t in every seed (Table S1)	no	no	no	yes

Table 4: Robustness of the main statements to the estimator and to the window of λ_∞. A statement is retained only if it holds with both estimators. Full table in Table S2.

Two statements pass this test: the `facsl` slope stays well below the diffusion–incision prediction, and V_{ov} does not reduce the spread of the spacing more than the storm time. The smaller residual spread of T_p at equal Σ^+ holds seed by seed with one estimator and one window only, and the hydrodynamic effects agree in direction but not in size or significance. The estimators differ mainly when the energy is spread over several modes, since λ_W is an arithmetic and λ_F a harmonic mean of the modal wavelengths (Sects. S1.4 and S1.5). This disagreement probably reflects patterns without a single dominant scale rather than an error in either estimator.

9.5 Limitations

- The morphological acceleration (`morfac` = 10) assumes that the bed evolves slowly compared with the flow. This may not hold during overwash, where the flow and the deposits interact strongly

- The periodic domain, with $L_y \approx 1200$ m, contains only a small integer number of channels, and the spacing evolves by discrete fusions. In the domain-size test, λ_∞ changes by about 15 % between $L_y = 1000$ and 2400 m (Sect. 5.6). Any effect of the domain on the absolute values is assumed to be common to all configurations.

- With three seeds per configuration, the intervals rely on a Student-t with 2 degrees of freedom and are wide, so several effects with λ_W may be real but unresolved. The seed noise ($\sigma_{\text{seed}} = 0.111$ in $\log \lambda_\infty$) is of the same order as several of the effects discussed.
- All configurations share a single wave-group chronology (`random` $= 0$), except along T_p, where each value has its own. The intervals therefore include the variability between seeds but not between chronologies, which Carrion-Bertran et al. (2026) found to be of the same order as the effects they studied.
- The fifteen runs that did not develop channels are excluded from the statistics, using a threshold on the spacing itself.
- The storm is idealized, with constant H_{m0}, T_p and z_{s0}, and a single initial profile is used. XBeach storm responses are also sensitive to the initial profile (Carrion-Bertran et al., 2026), and only one profile was tested.

10 Conclusions

In XBeach simulations of Castelldefels beach, the results do not support two of the three candidate mechanisms for the late spacing of the washover channels. Under the mapping $D \propto$ `facsl`, the diffusion–incision balance of Perron et al. (2009) predicts a positive exponent, whereas the measured one is negative with the Fourier estimator and compatible with zero with the wavelet estimator. The volume of water that overtops the crest does not account for the effect of any parameter better than the storm time.

The spacing increases with wave height, wave period and water level, which is consistent with the feedback between the overwash flow and the deposits described by Carrion-Bertran et al. (2026). This effect is significant only with the Fourier estimator, and it does not lead to a law for λ_∞. A direct test would require a measure of the incising capacity of the flow over the deposits rather than its cumulative volume. Future work could use the velocity field (u, v) and the suspended transport (`Susg`, `Svsg`) to check whether the flow develops its own alongshore scale before the bed does.

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